AI-Based Contract Risk Analysis and Clause Recommendation System

PHASE I REPORT

Submitted by

Nithiesh S	**230701215**
Muhammed Naveed Khan M J	**230701200**
Pranav Sriram	**230701235**
Pranesh Senthilkumar	**230701236**

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this Report titled "AI-BASED CONTRACT RISK ANALYSIS AND CLAUSE RECOMMENDATION SYSTEM" is the bonafide work of **NITHIESH S (230701215), MUHAMMED NAVEED KHAN M J (230701200), PRANAV SRIRAM (230701235), PRANESH SENTHILKUMAR (230701236)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

"This project addresses the following Sustainable Development Goal(s): Goal – SDG 16 : Peace, Justice and Strong Institutions."

SIGNATURE

Dr. J. Manoranjini,
Professor and Head of
Department of Computer Science
and Engineering,
Rajalakshmi Engineering
College,
Chennai – 602 105.

SIGNATURE

Dr. N. Srinivasan
Professor,
Department of Computer
Science and Engineering,
Rajalakshmi Engineering
College,
Chennai – 602 105.

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Internal Examiner

External Examiner

ABSTRACT

Contracts form the basis of almost all commercial relationships, but the manual examination of long legal documents is still a slow, expensive process and is prone to human error. Clauses that pose a risk—such as unlimited liability, automatic renewal, one-sided indemnity, or unclear terms regarding termination and payment—can lead organisations to face serious legal and financial repercussions if they are not identified. Currently available automated tools are mainly limited to keyword searching or clause-level classification, provide little scope for organisation-specific customisation, and seldom explain their decisions or suggest revised wording.

This paper outlines the design of an AI-Based Contract Risk Analysis and Clause Recommendation System which takes PDFs and Word documents as input, extracts and divides the clauses, classifies them into standard legal categories, assigns a risk severity on a four-level scale (Low, Medium, High, Critical), and uses a locally hosted Large Language Model served through Ollama to generate clause recommendations that are grounded in the context of the contract and organisational guidelines.

The proposed architecture integrates transformer-based clause classification (using LegalBERT/RoBERTa/DeBERTa models fine-tuned on the CUAD dataset) with a locally hosted Large Language Model served through Ollama for generating the recommendations, all made accessible via an interactive dashboard that provides risk visualisation, explanations linked to the source material, and exportable reports. The design is inspired by recent research into LLM-based legal risk assessment using Ollama and extends that work towards an end-to-end, explainable, organisation-adaptable process covering the entire pipeline from ingestion to recommendation.

6. CONCLUSION

6.1 CONCLUSION

This Phase I report presents the complete design and initial prototype implementation of an AI-Based Contract Risk Analysis and Clause Recommendation System. The system successfully combines modern NLP techniques, local Large Language Model via Ollama and Large Language Models to automate substantial portions of the contract review process.

The literature survey confirmed the relevance and novelty of an end-to-end, organisation-adaptable approach that goes beyond pure classification to include risk scoring and grounded recommendation generation. System analysis established the clear need for such a solution given the limitations of both manual review and existing automated tools. The modular architecture provides a strong and extensible foundation for further development.

The prototype demonstrated the practical feasibility of the full pipeline—from document ingestion and clause segmentation through classification, risk scoring, Ollama-based recommendation and recommendation generation—together with an interactive dashboard for result presentation. Successful completion of Phase I has laid a solid base for building a more advanced, fully evaluated and organisation-deployable system in the subsequent phase.

6.2 FUTURE WORK

The following enhancements are planned for future work:

- Support for multiple languages beyond English, including the adaptation of classification models and OCR pipelines.
- Real-time collaborative review features that allow multiple users to annotate clauses, discuss recommendations and record decisions.
- Continuous learning from user feedback so that accepted or rejected recommendations improve future risk scoring and generation quality.
- Advanced explainability features, including confidence scores and more detailed evidence linking for every risk verdict and recommendation.
- Domain-specific fine-tuning for industries such as healthcare, information technology, construction and financial services.
- Exploration of multi-agent architectures for more complex legal reasoning, negotiation support and automated redlining workflows.
- Rigorous quantitative evaluation on held-out CUAD splits and on organisation-specific annotated contracts, together with user studies measuring review time and decision quality.